

Math 1522 Calculus 2 Review
Fall 2026

1. Formulas to know

- (a) Derivative and graph of $\sin^{-1}(x)$.
- (b) State the Inverse Function Theorem.

2. Logarithmic differentiation

- (a) Find y' if $y = x^{\sqrt{x}}$.
- (b) Find y' if $y = (\sin x)^{\sin x}$.
- (c) Find $y'(0)$ if $y = \frac{(x+1)^4 e^x}{\sqrt{1+3x}}$.

3. Derivatives of logarithms

- (a) Find y if $y' = \frac{\sin(x)}{\cos(x)}$.
- (b) Find the domain and range of $f(x) = \ln(\sqrt{9 - e^x})$.
- (c) Let $f(x) = 3 - e^{-x}$. Find the domain and range of f , find $f^{-1}(x)$, and state the domain and range of f^{-1} . Does the graph of f have a horizontal asymptote?
- (d) Find $y'(1)$ if $y = 3^x \log_3(x)$.

4. Inverse trig functions

- (a) Prove the identity $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$ (ignore domain issues).
- (b) Find y if $y' = \frac{x^2}{\frac{1}{9} + x^6}$.
- (c) Find the equation of the tangent line to $y = 2 \tan^{-1}(x^2)$ at $x = 1$.
- (d) Write $\cos(\tan^{-1}(x/2))$ as an algebraic function of x .
- (e) Find y if $y' = \frac{x}{\sqrt{1-x^4}}$.

5. Hyperbolic functions

- (a) Find $\cosh^{-1}(2)$.
- (b) Evaluate $\lim_{x \rightarrow \infty} \coth x$.
- (c) Prove that the derivative of $\tanh x$ is $\text{sech}^2 x$.
- (d) Find y' if $y = \tan^{-1}(\sinh x)$, and simplify.
- (e) Evaluate $\int_{-\ln 3}^{\ln 3} \cosh x \, dx$.

6. Inverse Function Theorem

- (a) Show $f(x) = x + x^3 + 2x^5$ is 1-1, then find the derivative of f^{-1} at $(4, f^{-1}(4))$.
- (b) Show $f(x) = 5x + 2 \sin x + 2 \cos x$ is 1-1, then find the derivative of f^{-1} at $(2, f^{-1}(2))$.
- (c) Let $f(x) = x + e^x$. Find the equation of the tangent line to the graph of f^{-1} at $x = 1$.
- (d) Let $f(x) = 2x^3 + \ln x + \sin(\pi x)$ for $x > 0$ (assume f is 1-1). Find the slope of the tangent line to the graph of f^{-1} at $x = 2$.

7. L'Hôpital's rule

- (a) $\lim_{x \rightarrow 0^+} x^x$
- (b) $\lim_{x \rightarrow 0^+} (1 + \sin(2x))^{\cot x}$
- (c) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$
- (d) $\lim_{x \rightarrow 1^-} \frac{\ln x}{\cos^{-1}(x)}$
- (e) $\lim_{x \rightarrow \infty} x \left(\frac{\pi}{2} - \tan^{-1}(2x) \right)$
- (f) $\lim_{x \rightarrow \infty} \sin^{-1} \left(\frac{e^x}{2e^x + 1} \right)$
- (g) $\lim_{x \rightarrow \infty} \frac{\cosh(\ln x)}{x}$

8. Substitution

- (a) $\int_0^{(\pi/2)^{1/3}} x^2 \sin x^3 dx$
- (b) $\int_{\pi/4}^{\pi/2} \cot x dx$
- (c) $\int_1^e \frac{(\ln x)^2}{x} dx$
- (d) $\int \frac{\cos x}{4 + \sin^2 x} dx$
- (e) $\int \frac{dx}{\sqrt{x}(1 + \sqrt{x})}$

9. Integration by parts

- (a) $\int_1^e x^4 \ln(x) dx$
- (b) $\int_0^\pi e^x \sin(x) dx$
- (c) $\int x \sec^2(x) dx$
- (d) $\int x \sin(2x) dx$
- (e) $\int_0^1 x \tan^{-1}(x) dx$

10. Trig integrals

- (a) $\int \sin^3(x) \cos^2(x) dx$
- (b) $\int \tan^2(x) dx$
- (c) $\int \tan^2(x) \sec^4(x) dx$
- (d) $\int \sin(3x) \cos(2x) dx$
- (e) $\int_0^{\pi/3} \tan(x) \sec^3(x) dx$

11. Trig substitution

- (a) $\int_2^3 \frac{dx}{(x^2 - 1)^{3/2}}$
- (b) $\int \frac{dx}{(6 - 8x - x^2)^{3/2}}$
- (c) $\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{x^2 + 1}} dx$
- (d) $\int_{\sqrt{2}}^2 \frac{dx}{x^2 \sqrt{x^2 - 1}}$

(e) $\int_0^1 \frac{dx}{(1+x^2)^2}$

12. Partial fractions

(a) $\int \frac{3x^2+6x+2}{x^2+3x+2} dx$

(b) $\int \frac{x+4}{x^3-4x} dx$

(c) $\int \frac{3x^2-x+8}{x^3+4x} dx$

(d) Set up, but do not solve, the partial fraction decomposition of $\frac{x+1}{x^2(x-3)(x^2+1)^2}$.

13. Improper integrals

(a) $\int_0^1 \frac{\ln x}{x} dx$

(b) $\int_1^\infty \frac{dx}{x^p}$ (consider $p \leq 1$ and $p > 1$)

(c) $\int_0^\infty e^{x^2} dx$ (don't determine the value)

(d) $\int_0^\infty \frac{e^x}{(1+e^x)^3} dx$

(e) $\int_1^\infty \frac{\ln x}{x^3} dx$

(f) $\int_0^2 \frac{dx}{(x-1)^{2/3}}$

(g) $\int_2^\infty \frac{dx}{4+x^2}$

14. Numerical integration

(a) $\int_0^1 (x^3 - 3x^2 + 2x + 1) dx$: over or under estimates, then M_4 and T_4 .

(b) Use the trapezoid rule with $n = 3$ to approximate $\int_0^3 2^x dx$ in exact form.

(c) How large must n be so that the trapezoid rule approximates $\int_1^2 \frac{dx}{x}$ with error less than 10^{-4} ?

15. Sequences

(a) If an alternating sequence converges, what must it converge to?

(b) Write $\{-\frac{1}{8}, \frac{1}{27}, -\frac{1}{64}, \frac{1}{125}, \dots\}$ as a function of n .

(c) Monotonic? Bounded? $a_n = 1 + \frac{(-1)^n}{n}$ and $b_n = \frac{n}{n+1}$ ($n \geq 1$).

(d) Show that $\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$.

16. Test for Divergence (TFD)

(a) $\sum_{n=1}^\infty \frac{(-1)^n n}{n+1}$

(b) $\sum_{n=0}^\infty \frac{n^2+1}{6n^2+6n+1}$

(c) Can you use the TFD on $\sum_{n=2}^\infty \frac{1}{n^2}$?

(d) $\sum_{n=1}^\infty 2^{1/n}$

(e) Suppose $b_n > 0$ and $\sum b_n$ converges. What does the TFD say about $\sum \frac{1}{b_n}$?

17. Telescoping series

- (a) $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$
- (b) $\sum_{n=1}^{\infty} [\tan^{-1}(n+2) - \tan^{-1}(n)]$
- (c) $\sum_{n=1}^{\infty} [f(n) - f(n+1)]$, where $\lim_{n \rightarrow \infty} f(n) = L$
- (d) $\sum_{n=1}^{\infty} [f(n) - f(n+k)]$, k a positive integer, $\lim f(n) = L$
- (e) Determine if $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$ converges or diverges by writing out its partial sums.

18. Geometric series

- (a) $\sum_{n=0}^{\infty} \frac{4^{2n-1}}{17^n}$
- (b) $\sum_{n=-4}^{\infty} \left(\frac{1}{2}\right)^n$
- (c) $\sum_{n=1}^{\infty} \frac{3^{n+1}}{4^{n-1}}$
- (d) $2 - 1 + \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \dots$
- (e) For which a does $\sum_{k=1}^{\infty} \left(\frac{2a}{1+a^2}\right)^k$ converge? Find its value.

19. Integral Test

- (a) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$
- (b) Show $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $0 < p \leq 1$.
- (c) $\sum_{n=1}^{\infty} n e^{-n^2}$
- (d) Can you use the integral test on $\sum_{n=1}^{\infty} \frac{2+\sin n}{n^2}$?

20. p-series

- (a) $\sum_{n=5}^{\infty} \frac{1}{n^e}$
- (b) $\sum_{n=1}^{\infty} \frac{1}{n^{0.000001}}$
- (c) $\sum_{n=1}^{\infty} \frac{n^2 + \sqrt{n}}{n^4}$
- (d) For which p does $\sum_{n=1}^{\infty} \frac{1}{n^{2p-1}}$ converge?

21. Direct Comparison Test (DCT) and Limit Comparison Test (LCT)

- (a) $\sum_{n=1}^{\infty} \sin(1/n)$
- (b) $\sum_{n=0}^{\infty} \frac{\tan^{-1}(n)}{n^2+1}$
- (c) $\sum_{n=1}^{\infty} \frac{1}{2^n+n}$
- (d) $\sum_{n=1}^{\infty} \frac{n^2-2n}{n^{7/2}+1}$
- (e) $\sum_{n=1}^{\infty} \frac{1}{n^2+n \cos n}$

22. Alternating Series Test (AST)

- (a) $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$
- (b) $\sum_{i=1}^{\infty} \frac{\cos(\pi i)}{i^{0.1}}$
- (c) $\sum_{k=2}^{\infty} \frac{(-1)^{k+15}}{\ln k}$

- (d) $\sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}}$
- (e) $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{n+4}$
- (f) $a_n = \frac{1}{n}$ for n odd, $a_n = \frac{1}{n^2}$ for n even. Can you use the AST on $\sum (-1)^n a_n$?

23. Ratio Test

- (a) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$
- (b) $\sum_{n=0}^{\infty} \frac{(2n+1)!}{3^n}$
- (c) $\sum_{n=1}^{\infty} \frac{n^2 4^n}{n!}$

24. Root Test

- (a) $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1} \right)^n$
- (b) $\sum_{n=1}^{\infty} \frac{n^n}{2n^2}$
- (c) Show the Root Test is inconclusive for $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$. Which converges?
- (d) $\sum_{n=1}^{\infty} \frac{2^n}{n^5}$: Root Test? Ratio Test?

25. Absolute and conditional convergence

- (a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$
- (b) $\sum_{k=1}^{\infty} \frac{k \cos(\pi k)}{k^3+1}$
- (c) $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{2^n}$
- (d) $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$
- (e) Suppose $\sum a_n$ converges to 2. What is $\lim a_n$? Must $\sum |a_n|$ converge?

26. Radius of Convergence (ROC) and Interval of Convergence (IOC)

- (a) $\sum_{n=1}^{\infty} n^n x^n$
- (b) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{3^n (2n+1)}$
- (c) $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 4^n}$
- (d) Center, ROC, IOC: $\sum_{n=1}^{\infty} \frac{2^n (x-3)^n}{n^2}$

27. Power series from geometric series

- (a) Find a power series for $g(x) = \frac{1}{1+x}$ centered at $a = 1$. State the ROC and IOC.
- (b) Find a power series for $g(x) = \ln(x)$ centered at $a = 1$. State the ROC.
- (c) Find the function that $\sum_{n=2}^{\infty} 2n(n-1)x^n$ represents.
- (d) Find a power series for $f(x) = \frac{x}{(1-x^2)^2}$ centered at $a = 0$.
- (e) Find the exact value of $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

28. Taylor and Maclaurin series

- (a) Find the Maclaurin series for $\cosh(x)$ and $\sinh(x)$.

- (b) Degree 2 Taylor polynomial of $\sqrt{x}$ at $a = 9$.
- (c) Degree 3 Taylor polynomial of $\cos x$ at $a = \frac{\pi}{3}$.
- (d) Maclaurin series for $F(x) = \int_0^x e^{-t^2} dt$ and its ROC.
- (e) Find the exact value of $1 - \frac{\pi^2}{3^2 \cdot 2!} + \frac{\pi^4}{3^4 \cdot 4!} - \cdots$.
- (f) Find the exact value of $\sum_{n=0}^{\infty} \frac{(\ln 3)^n}{n!}$.
- (g) Find the exact value of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n 2^n}$.
- (h) Use a Maclaurin series to evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.

29. Approximations and error bounds

- (a) Approximate $\int_0^{0.1} \frac{dx}{1+x^4}$ to within 10^{-6} .
- (b) Use the degree 3 Maclaurin polynomial $T_3(x)$ to approximate $\tan^{-1}(x)$ near 0, and bound the error at $x = \frac{1}{2}$ with the Alternating Series Estimation Theorem (ASE).
- (c) Bound the same error with Taylor's Inequality, bounding $|f^{(4)}(x)|$ on $[0, \frac{1}{2}]$, where $f(x) = \tan^{-1}(x)$.
- (d) $f(x) = \sum_{n=2}^{\infty} 5n(n-1)x^n$: parts (i) to (vi).
- (e) How many terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ are needed to approximate its sum to within 0.001?
- (f) $f(x) = e^x$: find $T_3(x)$ and bound $|f(x) - T_3(x)|$ for $-1 \leq x \leq 1$.
- (g) Approximate $\int_0^1 e^{-x^2} dx$ to within 0.01 (use ASE).

30. Complex numbers

- (a) Compute $(1+i)^{10}$ in $a+bi$ form.
- (b) Compute $(i-1)^{12}$ in $a+bi$ form.

31. Differential equations

- (a) Show that $y = Ce^{x^2}$ solves $y' = 2xy$ for every constant C . Find the solution with $y(0) = 3$.
- (b) Sketch the direction field of $y' = x + y$ at the nine points with $x, y \in \{-1, 0, 1\}$. Along which line are the slopes 0? Sketch the solution through $(0, 0)$.
- (c) Find the equilibrium solutions of $y' = y^2 - 4$ and decide whether nearby solutions move toward or away from each.
- (d) Euler's method with $h = 0.5$ for $y' = x + y$, $y(0) = 1$, to approximate $y(1)$. Over or under estimate?
- (e) Euler's method with $h = -1$ for $y' = y - x$, $y(1) = 3$, to approximate $y(-1)$.
- (f) Solve $\frac{dy}{dx} = xy^2$, $y(0) = 1$. On what interval is the solution defined?
- (g) Solve $\frac{dy}{dx} = e^{x-y}$, $y(0) = \ln 2$.
- (h) Solve $\frac{dy}{dt} = 3y - 6$, $y(0) = 5$.
- (i) Bacteria grow at a rate proportional to size, from 200 cells to 600 in 2 hours. Find the number of cells $P(t)$ after t hours, and when $P = 5400$.
- (j) Logistic: $\frac{dP}{dt} = 0.5P \left(1 - \frac{P}{1000}\right)$, $P(0) = 200$. Carrying capacity, equilibria, growth, and $P(t)$.